Category Theory, Sheaf and Presheaf

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1 Category, Functor and Natural Transformation

$$
\begin{tikzcd}
A \arrow[r, "f"] \arrow[loop left, "1_A"] \arrow[dr, "g\circ f"'] & B \arrow[d, "g"] \arrow[loop right, "1_B"] \\
& C \arrow[loop below, "1_C"]
\end{tikzcd}
$$

Figure 1: Category

Category

Definition 1. A **category** $\mathcal{C}$ consists of the following data:

(i) A class $\mathrm{Ob}(\mathcal{C})$, called *objects* of category $\mathcal{C}$.

(ii) For a pair A and B of objects, a class $\mathrm{Mor}_{\mathcal{C}}(A, B)$[a] whose elements are called *morphisms* (or *arrows*) from A to B.

(iii) For a pair of morphisms f in $\mathrm{Mor}_{\mathcal{C}}(A, B)$ and g in $\mathrm{Mor}_{\mathcal{C}}(B, C)$, there is a *composition* $g \circ f$ in $\mathrm{Mor}_{\mathcal{C}}(A, C)$.

These data are to satisfy the following rules:

(i) For every object A, there is an identity morphisms 1_A in $\mathrm{Mor}_{\mathcal{C}}(A, A)$.

(ii) Composition is associative.

[a] Sometimes, we also denote it by $\mathrm{Hom}(A, B)$ or $\mathcal{C}(A, B)$.

Remark 1.

$$\begin{tikzcd} A \arrow[r, dashed, "f"] \arrow[dr, "g\circ f"'] & B \arrow[d, dashed, "g"] \\ & C \end{tikzcd}$$

Figure 2: Composition of Morphisms

$$\begin{tikzcd} A \arrow[loop above, "1_A"] \arrow[r, shift left, "f=f\circ 1_A"] & B \arrow[l, shift left, "g=1_A\circ g"] \arrow[loop below, "1_B"] \end{tikzcd}$$

Figure 3: Identity Morphisms

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[rr, bend left, "g\circ f"] \arrow[rrr, bend left=50, "h\circ(g\circ f)"] \arrow[rrr, bend right=50, "(h\circ g)\circ f"'] & B \arrow[r, "g"] \arrow[rr, bend right, "h\circ g"'] & C \arrow[r, "h"] & D \end{tikzcd}$$

Figure 4: Associativity of Composition

Functor

Definition 2. A **functor** $\mathbf{F} : \mathcal{C} \to \mathcal{D}$ between two categories $\mathcal{C}$ and $\mathcal{D}$ is given by the following data:

(i) A map $\mathbf{F} : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$

(ii) For every A and B in $\mathrm{Ob}(\mathcal{C})$, a map $\mathbf{F}$ given by

$$\begin{array}{rcccl} \mathbf{F} & : & \mathrm{Mor}_{\mathcal{C}}(A,B) & \longrightarrow & \mathrm{Mor}_{\mathcal{D}}(\mathbf{F}(A),\mathbf{F}(B)) \\ & & \phi & \longmapsto & \mathbf{F}(\phi) \end{array}$$

Remark 2.

$$\begin{tikzcd}[column sep=huge] A \arrow[r, dotted, bend left, "\mathbf{F}"] \arrow[d, "\phi"'] & \mathbf{F}(A) \arrow[d, "\mathbf{F}(\phi)"'] \\ B \arrow[r, dotted, bend right, "\mathbf{F}"'] & \mathbf{F}(B) \end{tikzcd}$$

Natural Transformation

Definition 3. Let $\mathbf{F}, \mathbf{G} : \mathcal{C} \to \mathcal{D}$ be functors. A **natural transformation** (or a **morphism of functors**) $\eta : \mathbf{F} \to \mathbf{G}$, is a collection $\{\eta_A\}_{A \in \mathrm{Ob}(\mathcal{C})}$ such that

(i) $\eta_A : \mathbf{F}(A) \to \mathbf{G}(A)$ is a morphism in the category $\mathcal{D}$, and

(ii) for each morphism $\phi : A \to B$ of $\mathcal{C}$ the following diagram is commutative

$$\begin{tikzcd}
\mathbf{F}(A) \arrow[r, "\eta_A"] \arrow[d, "\mathbf{F}(\phi)"'] & \mathbf{G}(A) \arrow[d, "\mathbf{G}(\phi)"] \\
\mathbf{F}(B) \arrow[r, "\eta_B"] & \mathbf{G}(B)
\end{tikzcd}$$

Sometimes we use the diagram

$$\begin{tikzcd}
\mathcal{C} \arrow[r, bend left, "\mathbf{F}"] \arrow[r, bend right, "\mathbf{G}"'] \arrow[r, phantom, "\Downarrow\eta"] & \mathcal{D}
\end{tikzcd}$$

to indicate that η is a morphism form $\mathbf{F}$ to $\mathbf{G}$.

Remark 3.

$$\begin{tikzcd}
A \arrow[r, dotted, bend left, "\mathbf{F}"] \arrow[rr, dotted, bend left=50, "\mathbf{G}"] \arrow[d, "\phi"', ""{name=P}] & \mathbf{F}(A) \arrow[r, "\eta_A"] \arrow[d, "\psi=\mathbf{F}(\phi)", ""'{name=Q}] & \mathbf{G}(A) \arrow[d, "\chi=\mathbf{G}(\phi)", ""'{name=R}] \\
B \arrow[r, dotted, bend right, "\mathbf{F}"'] \arrow[rr, dotted, bend right=50, "\mathbf{G}"'] & \mathbf{F}(B) \arrow[r, "\eta_B"] & \mathbf{G}(B)
\arrow[from=P, to=Q, dotted, "\mathbf{F}"]
\arrow[from=P, to=R, dotted, "\mathbf{G}"]
\end{tikzcd}$$

Example 1 (Determinant). TBA

2 Sheaf and Presheaf

Sheaf

Definition 4. TBA

Presheaf

Definition 5. TBA